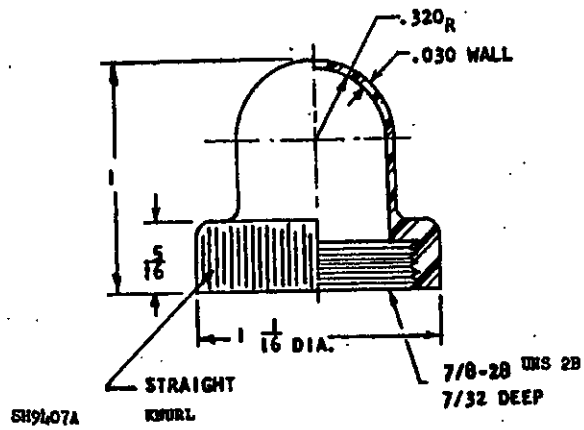
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MIL-L-3661/54A(SHIPS)
28 May 1971
SUPERSEDING
MIL-L-3661/54(SHIPS)
1 May 1968
(See page 2)

MILITARY SPECIFICATION SHEET

LENSES, INDICATOR LIGHT, STYLE LC40

The complete requirements for procuring the lenses described herein shall consist of this document and the latest issue of Specification MIL-L-3661.



INCHES	MM
.030	.76
7/32	5.56
5/16	7.94
.320	8.13
7/8	22.23
1	25.40
1-1/16	26.99

Figure 1 - Style LC40 lenses.

NOTES:
1. All dimensions are in inches. Unless otherwise specified, tolerances are ±1/64 inch (0.38mm) for fractions and ±0.005 inch (0.13mm) for decimals.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch =25.4 mm.

(A) Denotes changes.

FSC 6210

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REQUIREMENTS:

1. Qualification: Not required.
2. Material: Plastic, heat-resistant conforming to composition A, type I of MIL-M-20693.
3. Design and construction: See figure 1.
4. Knurl design: Straight.
5. The translucent white shall be such as to be readily recognized as white and that the light is "ON" when the lens is installed on an energized indicator light of any style (see table I).
6. The color of the red, yellow, green, and blue lens shall conform to those specified for category I (aviation lighting) by FED-STD-3.
7. The inside of the lens shall be frosted or pebbled so as to give a slight diffusing effect for all type designations except LC40TN3.
8. The combination of diffusion and colors shall be such that when a lens is installed on any energized mating indicator light specified in table I under an ambient of 50 foot-candles, a normal observer can readily tell, at a distance of 10 feet, whether the light is "ON", accurately distinguish the color and tell whether one or two lamps are lit.
9. Sealing: Dripproof.
10. Temperature cycling: Minus 55° to plus 125°C.
11. Random sampling from production runs shall be conducted after moisture content equilibrium (2-1/2 percent by weight) has been reached, and after coloring (if required). Inspection shall be conducted as follows:

    (a) Thread checks with standard "GO" and "NOT-GO" gauges conforming to Handbook H-28.
    (b) Dimensional checks:
        (1) Inner diameter of lens shall insure lamp clearance.
        (2) Inner diameter of flange shall insure contact pin switch operation.
    (c) Specific gravity check shall conform to MIL-M-20693.

12. Melting point property shall be tested and may be conducted as removed from mold.
13. When shock and vibration tests are conducted on the indicator light fixture, the lens shall remain intact during the test with no evidence of thread slippage.

Table I - Style LC40 lenses.

Type designation 1/	Lens material	Transmittance	Lens color	Superseding Data 3/ MS17128-	9000-S6202-74222
				Part No.	Piece No.
LC40CN3	Plastic	Transparent	Colorless	1	82
LC40WT3	Plastic	Translucent	White	2	81
LC40EN3	Plastic	Transparent	Red	3	77
LC40YN3	Plastic	Transparent	Yellow	4	80
LC40GN3	Plastic	Transparent	Green	5	79
LC40BN3	Plastic	Transparent	Blue	6	78
2/ LC40TN3	Plastic	Nondiffusing-unfrosted	Colorless	7	

1/ The style LC40 lenses shall mate with any of the indicator light housings shown on MIL-L-3661/61 through MIL-L-3661/65.

2/ Type designation LC40TN3 is the same as type designation LC40CN3 except the inside of the lens shall not be frosted or pebbled (see requirement 7).

3/ Supersedes piece 10 of Drawing 9000-S6202-73907.

QUALITY ASSURANCE PROVISIONS

First article inspection:

Complete lampholder assembly: When lenses specified herein are procured as a part of an assembly, the lens with the indicator light housing shall be subjected to the applicable tests listed in the qualification tests of MIL-L-3661.

Lenses: When lenses are procured separately, first article inspection shall consist of the following examination and tests:

Visual and dimensional examination
Sealing
Temperature cycling

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Quality conformance inspection: Quality conformance inspection shall consist of the following examination and test:

Visual and dimensional examination
Sealing

Sampling for quality conformance inspection: A random sample of lenses shall be selected from each lot in accordance with MIL-STD-105, inspection level I. The AQL shall be 2.5 percent defective. Any lens in the sample containing one or more defects or failing to conform to the sealing test shall be rejected. If the number of non-conforming lenses in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Preparing activity:
Navy - SH
(Project 6210-N311)

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